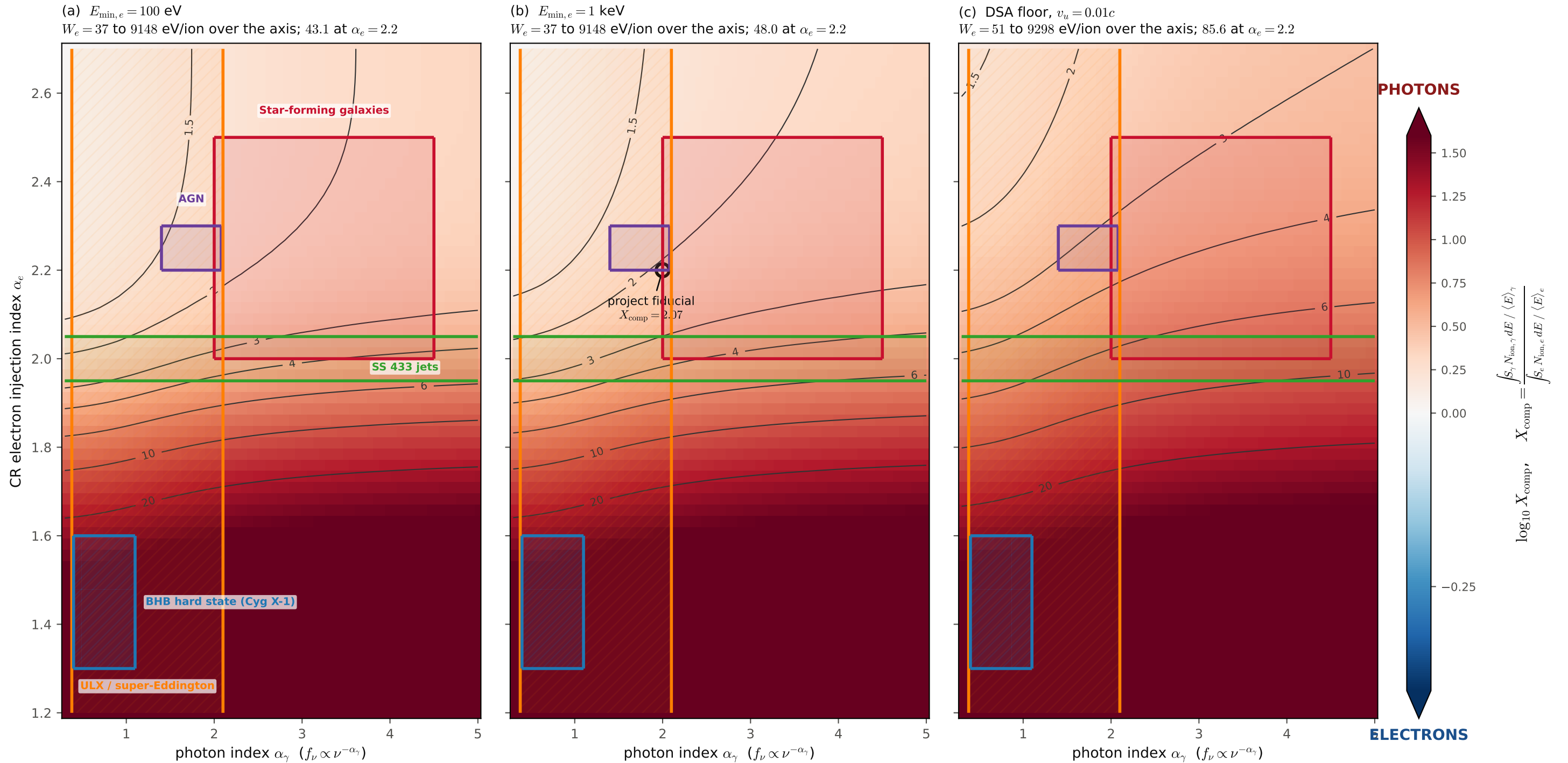




X_{comp} is the number of ionizations bought per unit energy injected, photons relative to electrons, at equal luminosity. $X_{\text{comp}} > 1$: photons are the cheaper ionizing agent.

SCENARIO IGM at $z=10$, $x_e = 10^{-4}$, pure H. Photon band [13.598, 1218] eV, from the HI threshold to the free-streaming onset $\tau_{\text{TGM}} = 1$. Electron band [$E_{\min,e}$, 10^{12}] eV.

BOXES span the range each index is given in the literature, on BOTH axes; all edges solid. A box drawn OPEN on an axis (no edges on that pair of sides, paler fill) has no measured index for it, so no range is invented.

HATCHED the photon index was fitted at 2--10 keV, which has ZERO overlap with our band; it is carried in by extrapolation, and in our band these objects are disc- not corona-dominated.

α_γ SOURCES Murchikova+20 (4.5 ± 0.4 over 13.6--54.4 eV, the only direct measurement in our band, but Galactic); Telfer+02 (AGN, 10.3--24.8 eV); Gladstone+09 (ULX and BHB, 2--10 keV).

α_e SOURCES Park+15 (DSA, $f(p) \propto p^{-3r/(r-1)}$, so $r=4$ gives exactly 2); Aharonian+24 (SS\,433, $\Gamma_e = 2$, measured in this exact convention).

CONVENTIONS CHECKED $L_\nu \propto \nu^{-\gamma} \Rightarrow \alpha = \gamma$; $N(E) \propto E^{-\Gamma} \Rightarrow \alpha = \Gamma - 1$; $f_\lambda \propto \lambda^\beta \Rightarrow \alpha = \beta + 2$, which is INVALID across the Lyman break --- hence Marques-Chaves+26's $\beta_{\text{UV}} = -2.88$ cannot be used here.

PANELS differ only in $E_{\min,e}$; (c) uses the DSA injection momentum $p_{\text{inj}} \simeq 3m_p v_u$ (Park+15), the lowest floor in the sourced range and so the most generous to electrons.